

Case Study: Simple Bookstore API (MongoDB

+ Node.js)

Create a **Bookstore** where you can:

1. Add books (title, author, price)
2. List all books
3. Find a book by title
4. Update book price

This teaches:

- Mongoose schemas
- CRUD operations (Create, Read, Update)
- Async/await usage

Step 1: Setup Project

```
mkdir bookstore
```

```
cd bookstore
```

```
npm init -y
```

```
npm i mongoose
```

Create bookstore.js file.

Step 2: Connect to MongoDB

Step 3: Define Schema and Model

Step 4: Insert a Book

Step 5: Fetch All Books

Step 6: Find a Book by Title

casestudymongo.js

```
const express = require('express');
```

```
const mongoose = require('mongoose');
```

```
const app = express();
```

```
app.use(express.json());
```

```
mongoose.connect('mongodb://127.0.0.1:27017/bookstore', {
```

```
  useNewUrlParser: true,
```

```
  useUnifiedTopology: true
```

```

    })

    .then(() => console.log("MongoDB connected"))

    .catch(err => {

        console.error("MongoDB connection error:", err);

        process.exit(1);

    });

const bookSchema = new mongoose.Schema({

    title: { type: String, required: true, index: true },

    author: { type: String, required: true },

    price: { type: Number, required: true }

}, { timestamps: true });

const Book = mongoose.model("Book", bookSchema);

async function seedData() {

    const count = await Book.countDocuments();

    if (count === 0) {

        await Book.insertMany([

            { title: "Clean Code", author: "Robert C. Martin", price: 499 },

            { title: "Eloquent JavaScript", author: "Marijn Haverbeke", price: 399 },

            { title: "You Don't Know JS", author: "Kyle Simpson", price: 299 }

        ]);

        console.log("Sample books inserted ");

    }

}

seedData();

app.get("/books", async (req, res) => {

    const books = await Book.find().lean();

    res.json(books);

});

```

```
app.post("/books", async (req, res) => {  
  const { title, author, price } = req.body;  
  const book = new Book({ title, author, price });  
  await book.save();  
  res.status(201).json(book);  
});  
  
app.listen(3000, () => console.log("Bookstore API running on http://localhost:3000"));
```